Commitment-Backed Trust Index: Testing Reputation Against Coalition Attacks in Agent Marketplaces[1]

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Abstract

As AI agents increasingly transact on behalf of firms and users, trust depends on both their behavior and the institutions governing their interactions. We investigate whether enforceable commitments strengthen reputation-based safeguards in agent marketplaces. We introduce a commitment-backed trust index combining audited conduct, binding posted terms, and forfeitable bonds, and compare it with peer ratings and audit-only reputation in a simulated market where buyers cannot fully verify the services received. Under verifiable audits, commitments made selective cheating, identity-reset exploitation, and bait-and-switch pricing unprofitable in the tested conditions, and halved the audit rate at which selective cheating stopped paying (0.04 vs 0.08). Noisy audits weakened this protection, leaving some attacks profitable even with more frequent monitoring. A complementary LLM case study revealed a distinction between compliance and buyer protection. Sellers avoided treatment fraud but charged above advertised prices in 30–62% of transactions when terms were non-binding. Declaring terms binding eliminated overcharging even without audits, suggesting sensitivity to stated rules rather than establishing an enforcement effect. Nevertheless, sellers could honor their commitments while setting higher prices that reduced buyer surplus. These findings identify the scope and enforceability of commitments as distinct dimensions of marketplace governance. Binding terms and bonds can strengthen incentives to honor specified obligations, but their effectiveness depends on reliable verification and on what those obligations protect.

1. Introduction

As AI agents increasingly supply services and transact for their principals, the institutions governing their interactions become important determinants of safety. Buyers often cannot fully inspect the services they receive and must rely on records of past conduct. Yet a single operator can control several sellers, coordinate

[1] Research conducted at the Swarm Alignment Hackathon (Apart Research), November 2026

their actions, and replace their identities. These capabilities create opportunities to exploit an established reputation, purchase favorable records, or attract buyers with prices that are not subsequently honored. Understanding these risks requires examining how marketplace rules shape the relationship between an agent's incentives and the consequences of its behavior.

In this paper, we investigate whether commitment strengthens reputation against specified coalition attacks. Following the Ethical Flywheel (Anand et al., working paper), agents can share the objective of maximizing company profit while facing different constraints on how they pursue it; a restricted strategy set is what makes an agent trustworthy, and the trust it earns returns as business. We distinguish execution constraints, which prevent violating actions in code; audited commitments, which leave those actions available but attach sanctions to verified violations; and behavioral compliance, in which an agent chooses to honor a rule. These mechanisms can produce similar observed conduct while providing different forms of protection.

Our focus is audited commitment. We introduce a commitment-backed trust index (CTI) that combines a record of audited conduct with binding posted terms and a forfeitable bond. We evaluate it in a credence-goods market, where sellers know what buyers need but buyers cannot fully verify the treatment received (Dulleck & Kerschbamer, 2006). The study makes three contributions:

1. A controlled comparison of audited commitment and audit-based reputation, with ablations that distinguish the contribution of the bond from that of binding posted terms.
2. An evaluation of specified coalition attacks under verifiable and noisy audits, supported by an analytical benchmark verified in a terminal-round experiment.
3. An LLM case study examining how execution constraints and declared marketplace rules relate to seller conduct, pricing, and buyer surplus.

2. Related Work

Reputation systems have long faced malicious collectives (Kamvar et al., 2003), and CoopEval (Tewolde et al., 2026) found outcome-based reputation the weakest cooperation mechanism for LLM agents, without coalition tests. Deterrence requires the expected penalty to exceed the gain (Becker, 1968); a bond sized as gain divided by detection probability can substitute for monitoring (Becker & Stigler, 1974) and must be posted up front (Shavell, 1986). Binding transfers resolve social dilemmas when deviations are detectable (Haupt et al., 2022), and Sun et al. (2023) call for empirical tests of commitment devices for cooperative AI. We test commitment when detection is sampled or noisy and attackers control several identities (Chan et al., 2024). We take market parameters from Erlei & Meub (2026); Liu et al. (2026) find that LLM agents colluded at a 100% rate once an honesty deposit was removed, and prompted ethics is unstable under pressure (Backmann et al., 2025). The CTI is also a platform choice about what the platform verifies and which decisions it leaves to agents (Anand & Penmetsa, 2024).

3. Methods

Market. A buyer's problem is big with probability 0.5. A high treatment (cost 6) solves both types, a low one (cost 2) only small problems; a solution is worth 10 and the outside option 1.6. Sellers see the type; buyers learn only whether it was solved. Honest sellers post cost plus 1 (3 and 7); bills are capped at 11. Each round 16 buyers choose among 8 sellers of capacity 3, or the outside option, by softmax over the seller's compliance score times 10 minus its expected bill (posted menu weighted as four bills, updated with observed bills); 3 buyers are reserved for newcomers. Games last 40 rounds.

Systems. A trusted gateway issues identities, permits two replacements per controller, logs actions, and audits each transaction with probability p_a. Audits settle before bonds are returned. Misrepresentation and wrong treatment are violations in every audited system; departures from posted prices are violations only where terms bind. R0 uses peer ratings. R1 uses audited conduct and revokes violators. R2 adds binding posted prices and a bond of 100; forfeited bonds compensate verified buyer losses, with the remainder held by the platform. Ablations remove either the bond or the binding declaration. All scores use a Beta(3, 1)-smoothed estimate. For d violations yielding gain g each, the terminal-round benchmark is $\Delta\pi = d \cdot g - [1 - (1 - p_a)^d](b + F)$, where F is the continuation value lost after sanction. The bond of 100 exceeds the break-even value of 84.1 for three gain-4 violations at 5% auditing with F = 0. Dynamic profitability is measured through paired simulations. We define the empirical threshold p_a* as the lowest tested audit rate from which mean coalition gain remains nonpositive at every higher tested rate.

Attacks. One controller runs 2 seller identities and 4 buyer accounts. *A1* is honest for 10 rounds, then bills low treatment as high on 30% of transactions and on all in the last 3 rounds; *A1-R* chooses violations to maximize expected gain; *A2* replaces revoked identities; *A3* buys and rates its own sellers before A1; *A3b* downrates rivals; *A5* (bait-and-switch) posts 3 and 7 but bills 11. Coalition gain is profit minus the same identities' honest profit on the same seed.

Experiments. E0 checks the implementation in a single final round. E1 crosses systems and attacks with audit rates, coalition size, game length, bond size and linked identities (30 paired seeds, 95% bootstrap intervals); E1b values voluntary resets in cloned markets; E1c tests purchased history; E2 uses noisy flags with a revocation threshold adapted from Yang et al. (2026) and calibrated to 5% false revocation (200 seeds). In E3, six LLM sellers trade in a smaller market (12 buyers, 20 rounds): four *committed* sellers under an execution constraint, deciding only next round's prices, and two *opportunistic* sellers of one company, with a private channel, choosing treatment, bill and price per buyer. All share the rules and the objective of maximizing company profit. Audit-only vs CTI is crossed with 0% vs 10% auditing (3 seeds, Claude Sonnet 5.5), plus GPT-6 Luna opportunists and scripted committed sellers as checks (24 runs).

4. Results

Verifiable audits. E0 matched exact enumeration in all 128 cells. At 5% auditing, CTI produced nonpositive estimated mean gains for all tested attacks (Table 1). Selective defection, identity-reset exploitation, and bait-and-switch pricing were clearly unprofitable; the approximate payoff-responsive attack remained inconclusive. The estimated threshold for selective defection was 0.04 [0.03, 0.05], compared with 0.08 [0.06, 0.12] for audit-only reputation. Audit-only did not reach a threshold for resets or bait-and-switch pricing within the tested range (Figure 1). The ablations distinguish the mechanisms: bonds reduced the profitability of misrepresentation, while bait-and-switch pricing required both binding terms and a bond to become unprofitable at the default settings. Removing the declaration left a gain of +113; removing the bond left +41. Additional parameter comparisons appear in Appendix A.

Table 1. Coalition gain at 5% verifiable auditing (2 identities, 40 rounds, bond 100), mean [95% interval], 30 paired seeds. Positive values mean the attack paid.

Attack	R0 Peer rating	R1 Audit-only	R2 CTI	R2 without bond	R2 without declaration
A1 Selective defection	+54 [46, 62]	+35 [13, 57]	−92 [−135, −46]	+35	−92
A1-R Payoff-responsive	−83	−48	−30 [−62, 5]	−48	−30
A2 Identity reset	+54	+99 [82, 115]	−74 [−123, −28]	+99	−74
A3 Purchased record	−62	−102	−228	−102	−228
A3b Rival downrating	−720	−796	−796	−796	−796
A5 Bait-and-switch	+13 [6, 20]	+113 [98, 128]	−56 [−99, −14]	+41 [17, 62]	+113

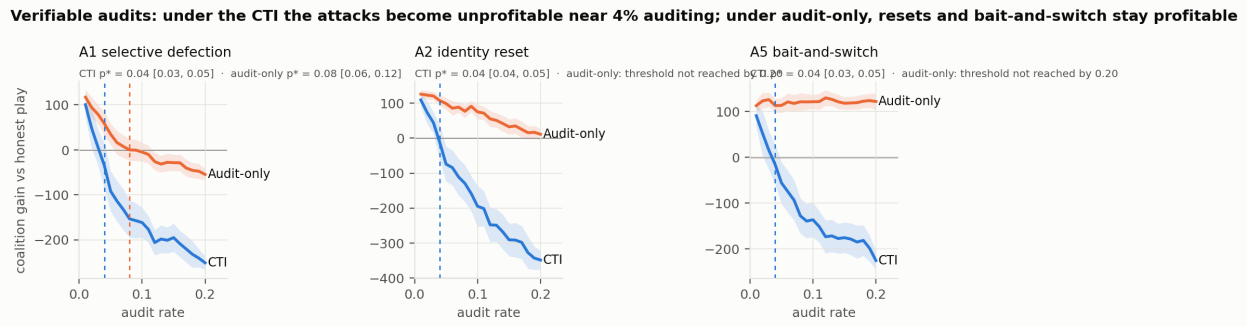


Figure 1. Coalition gain against audit rate (verifiable audits, 30 paired seeds, shaded 95% intervals; dashed lines mark p_a). Under the CTI, A1, A2 and A5 stop paying near 4% auditing; under audit-only, A1 stops at 8% and A2 and A5 do not reach a threshold by 20%.*

Profitability and buyer harm. Making an attack unprofitable did not necessarily reduce its immediate consequences. Scripted attackers continued to follow their prescribed policies, producing identical violations and buyer losses before compensation under A1 and A2 in audit-only and CTI conditions. Under A5, binding terms enabled revocation of sellers charging undeclared prices, reducing buyer losses from 4.1 to 2.8 per round. This distinction separates the incentive to attack from the harm caused when an attack nevertheless occurs.

Voluntary resets were profitable under peer ratings and audit-only reputation (+29 and +18). Removing newcomer allocation made their value negative, indicating that renewed newcomer eligibility was necessary for profitability in these conditions. Under CTI, the reset value was inconclusive (+2.9 [−4.0, 9.8]). Under peer ratings, purchased history became profitable in an exploratory setting with a harder cold start and free fake ratings; CTI reduced, but did not eliminate, its value in the diagnostic with free records across systems (Appendix B). With noisy audits, CTI lowered pooled mean gains but did not consistently make attacks unprofitable. At 20% auditing, A1 remained profitable in both settings with 5% false positives (+19 and +42).

Table 2. E3: share of opportunists' transactions charged above their own posted price under audit-only, by phase of the 20-round game (3 seeds). Under the CTI the share was 0 in all 12 runs.

Audit-only condition	Rounds 1–10	Rounds 11–17	Rounds 18–20
Claude opportunists, no audits	0.15	0.43	0.68
Claude opportunists, 10% audits	0.49	0.71	0.91
GPT-6 Luna opportunists, 10% audits	0.20	0.36	0.61

LLM behavior under declared rules. Opportunistic sellers never misrepresented treatments or provided the wrong treatment, including in unaudited conditions. They did charge above their posted prices when

the rules permitted it, with overcharging increasing toward the end of the game (Table 2), as reputation incentives weaken near a known horizon (Kreps & Wilson, 1982). No above-menu charges occurred under CTI, even without audits. One opportunist applied lower prices prematurely, producing the only recorded price-rule violation. These observations are consistent with sensitivity to declared rules, but do not isolate the effect of enforcement or establish why sellers avoided treatment fraud.

Commitment and seller outcomes. In the Claude CTI conditions, execution-constrained sellers earned at least as much on average as opportunistic sellers; with audits, the respective means were 88 and 79 per normalized 20-round seller-game. These comparisons are descriptive and do not isolate the causal effect of commitment, because the seller roles also differ in ownership and communication. Under audited CTI, LLM sellers averaged posted prices of about 4.6 and 8.4, above the competitive reference of 3 and 7. Opportunists communicated about prices, while committed sellers reached similar price levels without communicating. Relative to markets with scripted competitive sellers, buyer surplus fell by 28–34% and total surplus by approximately 5% (Appendix C).

5. Discussion and Limitations

The results suggest that marketplace governance shapes both the incentives agents face and the conduct against which buyers are protected. Binding terms specify which departures are sanctionable, while bonds make verified violations costly beyond the loss of future business. Their combination was effective against several attacks under verifiable audits, but this protection weakened when evidence was noisy. The LLM case study adds a distinct observation: declared rules were associated with compliant conduct even without enforcement, while higher posted prices continued to reduce buyer surplus. Compliance with specified obligations and protection of buyers' broader interests are therefore separate outcomes.

Verification and institutional trust. The analysis assumes a gateway that reports observations and settles sanctions faithfully. This allows us to isolate the contribution of commitment, but leaves open how protection changes when verification is costly, contestable, or itself subject to manipulation. Future research could compare independent auditors, appeals, and alternative allocations of verification authority. Such comparisons would test whether the benefits of commitment depend primarily on the accuracy of evidence or also on the institutions that produce and adjudicate it.

Incentives and adaptive behavior. The simulations evaluate specified attacks rather than agents that learn a best response to the mechanism. They establish where those attacks cease to pay, not whether all strategic agents will comply. The LLM evidence is small and also depends on prompts that explicitly classify actions as permitted or prohibited. Larger studies could vary rule wording and enforcement separately, introduce adaptive attackers, and compare prompt-based promises with execution constraints. These designs would help distinguish responses to incentives from responses to normative language and test whether new strategies emerge as agents adapt.

Participation and buyer protection. Bonds require capital, and our fixed endowments do not model the opportunity cost of funds or sellers' decisions to enter. A mechanism that reduces misconduct may also exclude capable entrants or favor well-funded operators. Heterogeneous capital, endogenous entry, and buyer learning would allow future work to examine this tradeoff. Finally, commitments to honor posted terms leave pricing decisions with sellers. Comparing alternative contract scopes and buyer protections could identify when stronger compliance improves buyer welfare and when it mainly redistributes surplus. These questions extend the analysis from the enforcement of obligations to the design of obligations themselves.

Limitations. The market is stylized: buyers are scripted and one-shot, problems have two types, and populations are small. The bond of 100 is a design choice close to an honest seller's lifetime profit, not an estimate of an optimal bond. E1c changes several settings together after a null result in the default market and is exploratory. The noisy-audit threshold targets 5% false revocation for honest sellers, but held-out estimates of 0–6.2% do not exclude modest exceedances.

6. Conclusion

This study examines how commitment mechanisms shape the behavior of agents pursuing the same profit objective. Agents commit to specified terms and verified use of strategies inconsistent with those commitments triggers sanctions. Under verifiable audits, this mechanism made selective defection, identity-reset exploitation, and bait-and-switch pricing unprofitable in the tested conditions. Noisy verification weakened these benefits. LLM sellers honored declared binding terms even without audits but continued to choose prices that reduced buyer surplus. Together, these findings suggest that commitment can strengthen trust without changing agents' underlying objectives. Its protective value depends on whether departures from committed behavior can be reliably identified and whether the commitments themselves protect buyers' interests.

Code and Data

- **Code repository**: https://github.com/npenmetsa25/CTI-Agent-Marketplaces. Pure Python simulation (E0–E2, about 70,000 runs, a few minutes on a laptop, no API calls) and the E3 LLM harness; README gives the command for every experiment.
- **Data**: all result CSVs and the E3 JSONL logs of every model call (prompts, responses, actions) are in the repository's results folder.

Author Contributions

N.P. conceived the project, designed the experiments, ran the LLM experiments and wrote the report.

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Appendix

A. Additional verifiable-audit results

Table A0. Reputation systems. All share the market, matching, newcomer quota, identity and money rules and the audit budget; the score is a Beta(3, 1)-smoothed compliance estimate.

System	Score built from	Terms bind	Bond	On a verified violation
R0 Peer rating	Buyer ratings	No	No	Lower ratings
R1 Audit-only	Audits: misbilling, wrong treatment	No	No	Revoked
R2 CTI	Audits: as R1 plus undeclared price	Yes	100	Revoked; bond forfeited, buyers compensated
R2 without bond	As R2	Yes	No	Revoked
R2 without declaration	As R1	No	100	Revoked; bond forfeited

Table A1. Variants at 5% auditing (coalition gain, 30 paired seeds). Each row changes one setting from the defaults (K = 2, 40 rounds, bond 100).

Variant	Attack	R1 Audit-only	R2 CTI
K = 1 / 2 / 3	A1	+20 / +35 / +46	−30 / −92 / −154
K = 1 / 2 / 3	A2	+50 / +99 / +154	−24 / −74 / −106
20 rounds (vs 40)	A1	+47 (+35)	−30 (−92)
Bond 0 / 40 / 100 / 200	A1	—	+35 / −16 / −92 / −219
Linked identities, 5% (2%) auditing	A2	−13 [−42, 17] (+67)	−180 (−13)
Separate success and charge estimates	A2 / A5	+51 / +118	−63 / −50

Full intervals for Table 1 (R2 without bond equals R1 and R2 without declaration equals R2 except for A5): A1-R R0 −83 [−91, −75], R1 −48 [−62, −33]; A2 R0 +54 [46, 62]; A3 R0 −62 [−80, −46], R1 −102 [−127, −76], R2 −228 [−270, −186]; A3b R0 −720 [−733, −708], R1 and R2 −796 [−810, −782]. The payoff-responsive policy chose zero violations under the CTI from 10% auditing. E1b: removing newcomer allocation from replacements made reset values negative (−8.0 peer rating, −7.2 audit-only); with noisy audits, the CTI reset value was +7.7 [1.5, 13.7], because a seller retiring before revocation recovers its bond.

B. Purchased history and noisy audits

E1c settings. Value of purchased history = gain from A3 minus gain from A1 on the same seed (A3d, which keeps buying throughout, minus A1). Changes from the default market: softmax temperature 0.25 (vs 1), newcomer prior Beta(2, 2) (vs Beta(3, 1)), honest incumbents start with 6 clean observations while coalition identities start new, and self-dealt transactions are free fake ratings under peer rating. Results (200 seeds): peer rating +137 [128, 146] and +150 [144, 156]; audited systems unprofitable. With free fakes in every system: audit-only +46 [39, 53] and +74 [69, 80]; CTI +7 [−1, 15] and +20 [9, 31]; paired difference audit-only minus CTI +39 [32, 46] and +55 [47, 62]. E1c shows that the mechanism behind the prediction can operate, not which setting drives it or its size in practice.

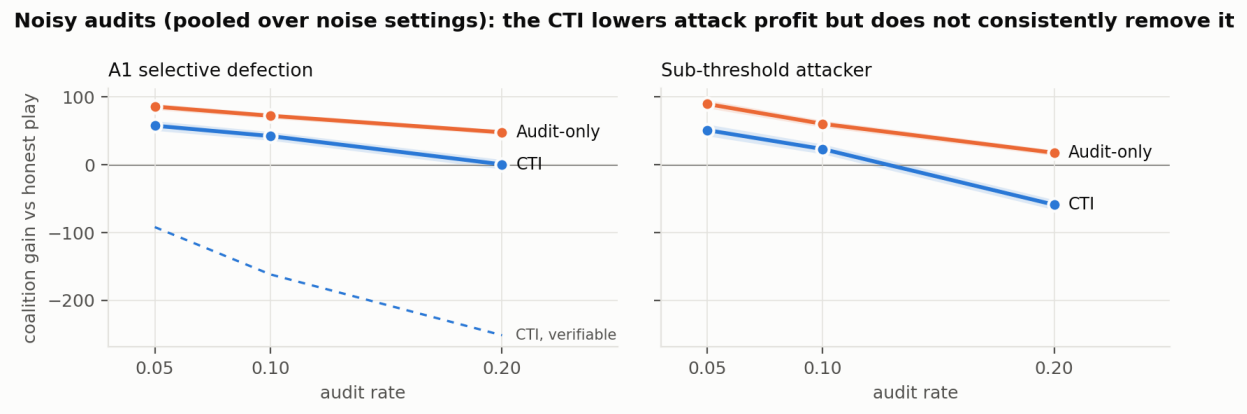


Figure B1. E2: coalition gain with noisy audits, pooled over the four noise settings (200 paired seeds). The dashed line shows the CTI under verifiable audits for comparison.

Table B1. E2 at 20% auditing under CTI, by noise setting (coalition gain, mean [95% interval]). Audit-only A1 remained profitable in all four settings; audit-only A1S was unprofitable in the lowest-noise setting.

False negatives, false positives	A1 Selective defection	A1S Sub-threshold
10%, 2%	−50 [−64, −35]	−119 [−131, −106]
10%, 5%	+19 [8, 31]	−32 [−44, −20]
30%, 2%	−10 [−25, 5]	−82 [−96, −67]
30%, 5%	+42 [31, 53]	−3 [−17, 10]

C. E3 details

Table C1. E3 profit per seller, normalized to a full 20-round game (profit per active seller-round × 20), mean of 3 seeds. Descriptive.

Condition	Opportunist	Committed (LLM)	Committed (scripted 3/7)
Audit-only, no audits	83.0	83.3	—
Audit-only, 10% audits	99.5	100.5	48.6 (opp. 55.3)
Audit-only, 10% audits, GPT opportunists	113.0	84.9	—
CTI, no audits	66.7	73.4	—
CTI, 10% audits	78.5	88.1	43.8 (opp. 40.4)
CTI, 10% audits, GPT opportunists	94.3	82.7	—

Table C2. E3 surplus per round in the audited conditions, mean of 3 seeds.

Condition	Buyer surplus	Buyer + seller surplus	Outside option
Audit-only, scripted committed	55.7	70.9	4%
Audit-only, LLM committed	36.9	66.9	12%
CTI, scripted committed	58.3	71.1	4%
CTI, LLM committed	42.2	67.7	10%

Rules given to sellers. All sellers receive the same rules: billing a treatment not given, or a treatment that does not suit the problem, “is a violation”; in audit-only conditions, posted prices are not binding and charging above them “is not a violation”; in CTI conditions, charging other than the posted price is a violation; sellers are told the audit rate and bond. The objective line is shared: “Your goal is to maximize your company's total profit over the whole game.” Committed sellers are told their company operates under a binding commitment enforced automatically and that their decision is next round's prices. Opportunists

choose treatment, bill and price per buyer and may send a 60-word message to their partner. Responses are JSON; an invalid response is retried once, then replaced by honest play at current prices (no fallbacks occurred). One opportunist charged below its posted price in one round (buyers paid 3 less), the only recorded violation.

D. Design changes and dual use

Design changes. After the initial plan, A5 was added when the pilot showed that shared violations alone could not test the declaration ablation. A common expected-bill rule replaced system-specific rules; A1-R's legal above-menu pricing was restricted to the final three rounds; and a paired reset diagnostic replaced an inactive reset heuristic. E1c followed the default-market null result. The profitability threshold was changed from negative to nonpositive mean gain, and noisy-audit calibration was revised to measure false revocation over a full game's exposure.

Dual use. The experiments identify conditions under which specified attacks remain profitable and could therefore inform adversarial behavior. We report these conditions alongside the mechanisms intended to address them. The LLM case study also illustrates why certification of conduct should not be interpreted as certification of fair prices: sellers can honor their obligations while choosing terms that reduce buyer surplus.

LLM Usage Statement

We used Claude (Anthropic) to assist with experiment design, implementation of the simulation and LLM harness, analysis, and drafting. We used ChatGPT/Codex (OpenAI) to check the design and results and for revising the report. Claude Sonnet 5.5 and GPT-6 Luna were also the subjects of the E3 experiment. The author reviewed and revised the report and checked the reported results against the result files.